Computer Science - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. ______________________ is the central node of 802.11 wireless operations.
   (a) WPA
   (b) Access Point
   (c) WAP
   (d) Access Port
2. How does a buffer overflow on the stack facilitate running attacker-injected code?
   (a) By overwriting the return address to point to the location of that code
   (b) By writing directly to the instruction pointer register the address of the code
   (c) By writing directly to %eax the address of the code
   (d) By changing the name of the running executable, stored on the stack
3. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
   (a) Generational
   (b) Blackbox
   (c) Whitebox
   (d) Mutation-based
4. Local caching of files is common in distributed file systems, but it has the disadvantage that
   (a) Temporary inconsistencies among views of a file by different machines can result
   (b) The file system is likely to be corrupted when a computer crashes
   (c) A much higher amount of network traffic results
   (d) Caching makes file migration impossible
5. A digital signature needs a
   (a) Private-key system
   (b) Shared-key system
   (c) Public-key system
   (d) All of them

True / False (10 marks)

Answer True or False

6. True or False: Encryption and decryption ensure both confidentiality and integrity of the data being transmitted.
7. True or False: The addition of the 8-bit two's-complement numbers 10000001 and 10101010 results in overflow.
8. True or False: A virus is a type of malware that operates solely as a standalone infected program without any hidden components.
9. True or False: All bit strings with an even number of zeros can be described with a regular expression.
10. True or False: The solution to the recurrence f(2 N + 1) = f(2 N) = f(N) + log N is $O((\log N)^2)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the nth digit in the proper fraction of two given numbers.
12. Write a function to group the 1 st elements on the basis of 2 nd elements in the given tuple list.

End of Paper